Pakistan Startup Ecosystem Analysis (2000–2018)

Comprehensive Data Analytics Portfolio Report & Ecosystem Insights

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433	3 Major Hubs	2016	49.9%
VALID STARTUPS	KARACHI, LAHORE, ISLAMABAD	PEAK FOUNDING YEAR	GLOBAL .COM SHARE

1. Executive Summary

This report evaluates the structural dynamics of the Pakistani startup ecosystem using census data spanning nearly two decades (2000–2018). By combining Python-driven data cleaning, feature engineering, and text mining with an interactive Power BI dashboard, this study measures urban startup density, temporal growth velocity, sector saturation, and web domain strategies.

2. Data Wrangling & Cleaning Methodology

The raw dataset contained 553 initial entries. Rigorous preprocessing was applied:

- **Null Elimination:** Purged 120 entirely empty records, retaining 433 validated startup profiles.
- **Location Standardization:** Applied string parsing to unify urban variations into structured hubs (Karachi, Lahore, Islamabad, and secondary regions).
- **Temporal Extraction:** Utilized Regular Expressions (`r'\b(19\d\d|20\d\d)\b'`) to extract numeric founding years from unstructured date strings.

3. Key Analytical Findings & Insights

Geographic Hub Concentration

Karachi anchors the ecosystem with **46.2%** of startups, followed by Lahore (**28.2%**) and Islamabad (**12.2%**). Together, the Big Three concentrate over **86%** of national startup activity.

Growth Velocity & Peak Era

Startup creation surged exponentially between 2014 and 2017 following mobile broadband rollouts, hitting an all-time high in **2016** with **104 newly founded startups**.

4. Power BI Executive Dashboard Architecture

Designed as a single-page interactive control center featuring KPI cards, dynamic donut charts, founding trend lines, and sector breakdowns:

